Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2010-0318	Issued	2010-09-13

UT Thickness Survey - Regenerator Overhead and 4-RA-104 New Spool Baseline

Scope and Responsibilities

This report details the ultrasonic thickness survey conducted at the Hamilton Plant within the ATU-100. The scope includes the 3-year interval inspection of the regenerator overhead circuit and the baseline measurement of a newly installed spool. Kevin Doucette served as the lead UT technician for these activities, ensuring all data collection adhered to the site-specific requirements for the Amine Treating Unit.

Halward Inspection Services Ltd. performed the measurements under HIS-UT-002 Rev 1, which has been accepted for use at this facility. The inspection was executed in accordance with QSCP-NDT-001 Rev 1, utilizing a Krautkramer DMS 2 0317. The instrument's accuracy was verified per HIS-CAL-10-087, dated 2010-09-13, to ensure compliance with ASTM E797 and API 570 requirements for digital thickness measurement.

Regenerator Overhead Circuit Survey

The survey covered lines 3-RV-101 and 2-RV-102, both constructed of A106 Gr.B material. Line 3-RV-101 has an OD of 88.9 mm and a nominal thickness of 7.62 mm, while 2-RV-102 has an OD of 60.3 mm and a nominal thickness of 5.54 mm. Both lines are designed for 345 kPa(g) and 121 °C. Per QSCP-IP-ATU100 Rev 1, the inspection interval for this circuit is set at 3 years due to its classification and service history.

Data was compared against the previous survey recorded in HIS-2007-0388. Following the requirements of QSCP-NDT-001 Rev 1, scanning was performed at all elbows and tees rather than spot measurements. The 138 CMLs identified in QSCP-IP-ATU100 Rev 1 were evaluated against the retirement thicknesses of 3.56 mm for 3-RV-101 and 2.79 mm for 2-RV-102. All readings remained above these t-min values, and no NCRs were issued during this cycle.

4-RA-104 Spool Baseline

A baseline survey was conducted for the new spool on line 4-RA-104, which was replaced in August 2010. This replacement was managed under QSCP-MOC-2010-022, which specified an in-kind replacement of the rich-side exchanger piping with an immersion-grade coating and revised insulation. The line is A106 Gr.B with an OD of 114.3 mm and a nominal thickness of 8.56 mm, designed for 862 kPa(g) and 127 °C.

Baseline readings were established at CMLs -03 to -05 as required by the MOC. These initial values will serve as the reference for future short-term and long-term corrosion rate calculations. The installation and materials were verified to meet the design requirements of ASME B31.3. The retirement thickness for this section is established at 3.81 mm.

Corrosion Rate and Remaining Life Analysis

Corrosion rates were calculated by Kevin Doucette using the larger of the long-term rate and the short-term rate, with the latter utilizing the 3-year data spread between HIS-2007-0388 and the current survey. Remaining life was determined by subtracting the t-min from the current thickness and dividing by the governing rate. This analysis ensures the piping remains within the safety margins defined by API 570.

Based on the remaining life calculations and the Class 1 designation for the regenerator overhead, the next thickness survey is scheduled for 2013. This maintains the 3-year interval specified in QSCP-IP-ATU100 Rev 1. The 4-RA-104 spool will follow the standard Class 2 interval for amine service, with the next inspection due within 10 years or half the remaining life, whichever is sooner.

Thickness readings, measured 2010-09-13

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2007-09-18	6.95	6.78	0.056	0.044	>50	2013-09-13	
3-RV-101-02	7.62	3.56	2007-09-18	6.89	6.59	0.068	0.075	40.6	2013-09-13	
3-RV-101-03	7.62	3.56	2007-09-18	6.56	6.16	0.094	0.090	27.8	2013-09-13	
3-RV-101-04	7.62	3.56	2007-09-18	6.66	6.43	0.080	0.048	35.9	2013-09-13	
3-RV-101-05	7.62	3.56	2007-09-18	6.87	6.63	0.072	0.084	36.8	2013-09-13	
3-RV-101-06	7.62	3.56	2007-09-18	6.71	6.43	0.080	0.068	35.9	2013-09-13	
3-RV-101-07	7.62	3.56	—	—	6.20	—	—	—	2013-09-13	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-08	7.62	3.56	—	—	6.71	—	—	—	2013-09-13	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2007-09-18	4.83	4.55	0.068	0.066	26.1	2013-09-13	
2-RV-102-02	5.54	2.79	2007-09-18	4.67	4.60	0.068	0.053	26.5	2013-09-13	
2-RV-102-03	5.54	2.79	2007-09-18	5.01	4.84	0.044	0.066	31.1	2013-09-13	
2-RV-102-04	5.54	2.79	2007-09-18	5.12	4.84	0.051	0.051	40.2	2013-09-13	
2-RV-102-05	5.54	2.79	—	—	5.04	—	—	—	2013-09-13	
2-RV-102-06	5.54	2.79	—	—	4.82	—	—	—	2013-09-13	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-03	8.56	3.81	—	—	8.59	—	—	—	2020-09-13	
4-RA-104-04	8.56	3.81	—	—	8.52	—	—	—	2020-09-13	
4-RA-104-05	8.56	3.81	—	—	8.43	—	—	—	2020-09-13	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2010-09-13	_____